(Please write your Enrollment Number) Enrollment No. 15101172022

MID-TERM EXAMINATION
(B.Tech IT/ CSE/ MAE/ ECE/ DMAM,/ CSE-AI/ ECE-AI)
(Semester 3rd)
(October, 2023) OFF LINE mode

Subject Code: BCS 201	**Subject:** Data Structures
Time : 1 ½Hours	**Maximum Marks :** 30
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**
	(a) Imagine a case of modified STACK implementation which supports an operation like REVERSE, which reverses the order of elements on the STACK, in addition to the PUSH and POP operations. Prove that by using modified STACK implementation a queue can be implemented where ENQUEUE takes a sequence of three operations of STACK and DEQUEUE takes a single operation of STACK.	

	(b) Consider the following C function definition
	int Mystery (int a, int b, int c){
	if ((a >= b) && (c < b) return b;
	else if (a >= b) return ~~Trial~~ Mystery (a,c,b);
	else return Mystery (b,a,c); }
	What output is expected from Mystery? Justify the answer.	

	(c) If an array having N elements, what is the maximum number of recursive calls made by the binary search algorithm? Justify your answer.	
	(d) How array and link list are different on the basis of time and space complexity?	
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**
	(a) Consider an array A[L_1:U_1, L_2:U_2,……,Ln:Un] of n dimensions where Li are the lower bounds and Ui are the upper bounds and 'E' is the size of each array element. The array will be stored in row major order beginning at address 'B' then find the address of element A[S_1,S_2,…..,Sn].	
	(b) Write an algorithm to traversing the single link list in reverse order. Reverse order traversal is scanning of nodes from last node to first node.	
	(c) Implement queue using array and point out the problem with this data structure. Also propose the solution.	
Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**
	(a) Write an algorithm to delete the Kth element from the end of a given single linked list.	
	(b) By using a Stack convert the given infix expression $A + B \times \log C \uparrow D \uparrow E \div F - G$ into postfix expression and prefix expression. Where symbol " ↑ " is a power operator. Show each step and justify the logic.	
	(c) Write an algorithm to print the frequency of each element found in a given Circular Link List.	

while (temp! =